

(1)

Joint Density of states

In semiconductors, the joint density of states (JDOS) characterises the combined density of available states for energy of the charge carriers.

From the
Suppose an electron is in E_1 state of valence band and upon the absorption of energy $h\nu$, it went to

Energy of the electron in having energy conduction band of state E_2

$$E_2 = E_c + \frac{\hbar^2 k^2}{2m_c} \quad - (1)$$

Energy of the electron in valence band of state E_1 having energy E_1

$$E_1 = E_v - \frac{\hbar^2 k^2}{2m_v} \quad - (2)$$

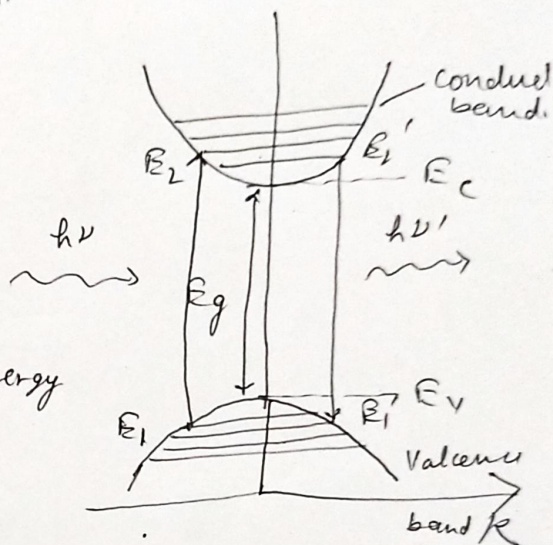
Subtracting (2) from (1) we have

$$E_2 - E_1 = (E_c - E_v) + \frac{\hbar^2 k^2}{2} \left[\frac{1}{m_c} + \frac{1}{m_v} \right]$$

$$\text{Suppose } \frac{1}{m_r} = \frac{1}{m_c} + \frac{1}{m_v}$$

$$E_2 - E_1 = E_g + \frac{\hbar^2 k^2}{2m_r} \quad - (3)$$

$$\text{where } E_g = E_c - E_v$$



(2)

~~E₂~~ E₂

~~Since~~ Since $E_2 - E_1 = h\nu$ — (4)

Now equation (3) can be written as.

$$(h\nu - E_g) = \frac{\hbar^2 k^2}{2m_r}$$

$$\Rightarrow k^2 = \frac{2m_r}{\hbar^2} (h\nu - E_g) \text{ — (5)}$$

substituting (5) in (1)

$$E_2 = E_c + \frac{\hbar^2}{2m_c} \frac{2m_r}{\hbar^2} (h\nu - E_g)$$

$$E_2 = E_c + \frac{m_r}{m_c} (h\nu - E_g) \text{ — (6)}$$

Now

$$E_2 \propto \nu$$

$$f(E_2) dE_2 = f(\nu) d\nu \text{ — (7)}$$

$$\frac{dE_2}{d\nu} = \frac{m_r h}{m_c}$$

$$f(\nu) = \frac{\pi}{2} \left(\frac{8m_c}{h^2} \right)^{3/2} (E_2 - E_c)^{1/2} \frac{dE_2}{d\nu}$$

$$\begin{aligned} \Rightarrow f(\nu) &= \frac{\pi}{2} 8 \left(\frac{2m_c}{h^2} \right)^{3/2} (E_2 - E_c)^{1/2} \frac{m_r h}{m_c} \\ &= \frac{\pi}{2} 8 \left(\frac{2m_c}{h^2} \right)^{3/2} \left(\frac{m_r}{m_c} \right)^{1/2} (h\nu - E_g)^{1/2} \frac{m_r h}{m_c} \end{aligned}$$

(3)

$$g(\nu) = 4\pi \left(\frac{2m_e}{h^2} \right)^{3/2} \left(\frac{m_r}{m_e} \right)^{3/2} h (\hbar\nu - E_g)^{1/2}$$

$$g(\nu) = \frac{4\pi}{h^2} (2m_r)^{3/2} (\hbar\nu - E_g)^{1/2}$$

This is the expression for optical joint density of states.